# PROGRAM ANALYSIS AND PROJECT PLANNING FOR A SMART PARKING SYSTEM

## Introduction

The rapid growth of employee strength in technology parks has led to severe parking congestion issues. At InfraTech Park, employees often spend up to 30 minutes searching for available parking spots, resulting in frustration, loss of productivity, and traffic congestion near entry points. Management has mandated the development of a Smart Parking System that efficiently guides drivers to available parking spaces using existing infrastructure only, within strict budget and time constraints.

This document presents a comprehensive project plan covering problem analysis, requirements, scope, feasibility, cost analysis, schedule, and risk management. The plan strictly adheres to: - A budget cap of USD 5,000 for the planning phase - A 2-week timeline for completion - No new hardware purchases, relying solely on existing low-resolution cameras and barrier gates

## Problem Statement & SMART Objectives

InfraTech Park faces chronic parking congestion due to the absence of real-time visibility into parking space availability. Drivers randomly search for spots, increasing traffic, fuel consumption, and employee dissatisfaction. The lack of an automated guidance mechanism leads to inefficient utilization of parking capacity.

The problem is to design a cost-effective, camera-based smart parking solution that detects parking availability and guides drivers efficiently using existing infrastructure.

## Objectives

- Reduce parking search time by at least 40% within 3 months of deployment by providing real- time availability guidance.
- Achieve ≥85% accuracy in detecting occupied vs. vacant parking zones using existing low- resolution cameras.
- Deliver a functional prototype design and plan within 2 weeks, ready for management approval.
- Operate entirely within the USD 5,000 budget during the planning and feasibility phase.
- Ensure system scalability, allowing future integration of mobile apps and advanced sensors without redesign.

## Scope Statement

### In-Scope Features (Initial Release)

The following features are included in the first release:

**Camera-Based Vehicle Detection**

- Use of existing entry/exit cameras to count vehicles entering and leaving.
- Zone-level occupancy estimation (not individual slot-level).

**Parking Availability Estimation Engine**

- Rule-based and lightweight computer vision algorithms.
- Occupancy calculation based on total capacity minus detected vehicles.
- Driver Guidance System

**Admin Dashboard (Web-Based)**

- Real-time occupancy view
- Historical data (daily trends)
- Manual override for security/admin staff

**System Logging & Reports**

- Daily utilization reports
- Peak-hour congestion reports

### Out-of-Scope Features (Explicitly Excluded)

The following are excluded from the initial scope:

- Mobile application or mobile-based booking
- Individual parking slot detection
- Automated payment or billing integration
- License plate recognition
- High-resolution live video streaming
- New hardware (ground sensors, LiDAR, RFID, ANPR cameras)
- AI-based deep learning models requiring GPU infrastructure

## Functional and Non-Functional Requirements

| ID | Requirement Description |
|---|---|
| **FR1** | The system shall detect vehicles entering and exiting using existing cameras |
| **FR2** | The system shall calculate total available parking spaces per zone |
| **FR3** | The system shall update parking availability in near real-time (≤10 sec delay) |
| **FR4** | The system shall display availability status at entry points |
| **FR5** | The system shall provide an admin web dashboard |
| **FR6** | The system shall generate daily and weekly utilization reports |

## Non-Functional Requirements

| CATEGORY | REQUIREMENT |
|---|---|
| Performance | Availability updates within 10 seconds |
| Accuracy | >=85%detection accuracy |
| Reliability | System uptime >=95% |
| Security | Role based access for admin dashboard |
| Usability | Simple,text-based UI Drivers |

## Technical Feasibility Analysis

### Use of Existing Cameras

Given the limitation of older, low-resolution

- Use vehicle counting at entry/exit points instead of slot-level detection
- Apply background subtraction and motion detection algorithms
- Use open-source libraries such as OpenCV
- Prefer zone-level occupancy for higher accuracy

This approach minimizes computational complexity and avoids expensive AI models.

## System Architecture (High-Level)

- Camera Feed Input
- Vehicle Detection Module
- Occupancy Calculation Engine
- Central Server (Cloud/Local VM)
- Display Boards & Admin Dashboard

## Work Breakdown Structure (WBS)

- Requirement Analysis
- Feasibility Study
- System Architecture Design
- Algorithm Selection & Validation
- Cost Analysis

| **Day** | **Activity** |
|---|---|
| Day 1-2 | Requirement Analysis &Problem Study |
| Day 3-4 | Feasibility &camera Assessment |
| Day 5-6 | Architecture &Design |
| Day 7-8 | Algorithm Validation |
| Day 9-10 | Cost& Risk Analysis |
| Day 11-12 | Draft Documentation |
| Day 13 | Review& Refinement |
| Day 14 | Final Submission |

## Initial Risk Register

| Risk Id | Risk Description | Impact | Mitigation Strategy |
|---|---|---|---|
| R1 | Low Camera quality affects accuracy | High | Use zone level detection |
| R2 | API trial limits exceed | Medium | Prefer Open cv-based Processing |
| R3 | Lighting/weather conditions | Medium | Adaptive thresholding |
| R4 | Network Latency | Low | Local processing where possible |

## Code and budget Analysis

| Item | Cost |
|---|---|
| Cloud VM(2 months trial &buffer) | 800 |
| OpenCv & Open-Source Tools | 0 |
| Cloud Vision API | 500 |
| Web Dashboard Development | 1,200 |
| Computer Vision Development | 1,500 |
| Testing & Validation | 400 |
| Documentation &Reporting | 300 |
| Contigency Buffer | 300 |
| Total | 5,00 |

## Efficiency & Future Enhancements

- Lightweight algorithms instead of deep learning
- Minimal cloud dependency
- Zone-based logic for robustness

## Future Enhancements (Post Phase)

- Mobile app integration
- Slot-level detection
- License plate recognition
- IoT sensors when budget allows

## Conclusion

This project plan delivers a feasible, cost-effective, and efficient Smart Parking System within the given constraints. By leveraging existing cameras, open-source tools, and zone-based detection, the proposed solution significantly reduces parking congestion while staying within the USD 5,000 budget and 2-week timeline. The plan also ensures scalability for future smart infrastructure upgrades.